

Review article

Challenges and opportunities of using meat inspection data to improve pig health traits by breeding: A review

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ABSTRACT

Pig breeding associations focus on breeding healthier pigs; however, valuable health-related phenotypes are limited. Apparent organ lesions documented during meat inspection may provide an opportunity to integrate phenotypes into the breeding program. Nevertheless, several factors influence the scoring of organ lesions causing intra-abattoir and inter-abattoir variations in the estimated prevalence. The utilization of recorded lesions as health-related phenotypes depends on a precise and uniform lesion definition and standardized data recording but these conditions are currently not sufficiently met in meat inspection data. Meat inspector training is often quoted as producing a sensitivity-increasing approach in the literature, and statistical approaches such as data scaling or latent-class models might handle this systematic abattoir effect. However, initial heritabilities have been estimated to demonstrate the potential of meat inspection data. Improving pig health via breeding by using this data currently requires further efforts but seems to be a valuable method for the future.

1. Introduction

During the last decade, consumers and current policy have increasingly demanded improvement in animal welfare and health (Merks et al., 2012), and according to Gamborg and Sandøe (2005) animal health is one of several main values in sustainable farm animal breeding. This is why pig breeding associations focus on functional traits and disease resistance in breeding programs (Mellencamp et al., 2008; Merks et al., 2012). However, phenotypes that represent animal health and welfare are limited, as they are expensive and difficult to obtain for a large number of pigs (Gourdine et al., 2010; Merks et al., 2012). Gourdine et al. (2010) reported contrary effects among some traits in their simulation study by showing a decrease in growth rate and the leanness of pigs raised for slaughter when selecting for mother ability and sow longevity as welfare breeding goals. Several studies have investigated whether meat inspection data could be used for health surveillance purposes (Correia-Gomes et al., 2016; Knage-Rasmussen et al., 2015; Teixeira and Boyle, 2014). As organ lesions are constantly documented during meat inspection as part of food safety regulation (EC No. 854/2004, 2004), this data pool is potentially valuable as phenotypes for breeding based on health traits. When including new traits in a breeding goal, it should generally be examined if the following aspects have been given due prominence: precise trait definition (validity), proper data recording (reliability), sufficient

heritability, absence of negative correlation with other relevant breeding traits and economic importance (Wachendorf et al., 2018). Considering these requirements, meat inspection data have been identified in literature as possessing major disadvantages, questioning their usability for breeding purposes (Eckhardt et al., 2009; Hoischen-Taubner et al., 2011; Steinmann et al., 2014).

The focus of this review therefore is on the identification of the opportunities and limitations of meat inspection data for the purpose of breeding towards improved pig health. For this purpose, an overview of the current status of meat inspection is given first, and subsequently challenges and opportunities associated with abattoir data are identified.

2. Methods

A systematic search of scientific literature was carried out, using the scientific research engines PubMed, Web of Science and ScienceDirect (www.ncbi.nlm.nih.gov/pubmed/; www.webofknowledge.com; www.sciencedirect.com), visited last time on May 9th, 2018. A comprehensive list of key terms was used to represent the aim of this review: meat inspection/abattoir data, pig health surveillance and breeding pig health. The keywords were extended by further search terms (e.g., reliability, validity, prevalence and evaluation of meat inspection/abattoir data, abattoir data as health monitoring, meat inspection data for

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breeding purposes). Since legislation, regulations and practice have changed during the last decade, focus was on studies published since 2008. After retrieving a comprehensive list of abstracts, any studies appearing to deal with the key and extended terms were obtained and reviewed in full. The majority of the selected papers have been published in peer-reviewed journals. All type of studies (e.g. original field investigations, simulation studies, review articles) were included in the selection. The literature search was also extended to secondary publications (i.e. regulations, reference books and publications of symposiums).

3. Meat inspection data

3.1. Recording of meat inspection data

The purpose of meat inspection is to examine products of animal origin for human consumption and to monitor specific hazards, contaminants and residues of veterinary drugs. The inspection tasks comprise food chain information, laboratory testing, specified risk material, by-products, animal welfare, ante-mortem and post-mortem inspection (EFSA, 2011). EU Directive (EC No. 854/2004, 2004) lays down a basic structure for the organization of the meat inspection process and is implemented by different guidelines at national level, which allow for variations between EU countries. Such variations were discussed by Harley et al. (2012), who pointed out the main differences in the structure and training requirements of meat inspection professionals. For example, there are differences in employment-status and qualification of official veterinarians and of official auxiliaries, as well as variations in the ratio of veterinarians to official auxiliaries, which may affect employee ability and/or working morale (Stärk et al., 2014). In addition, EC Directive (EC No. 854/2004, 2004) regulates that post-mortem inspection in pigs can be carried out by inspecting only some of the organs, as listed in the legislation. The organ lesions for which registration is mandatory during post-mortem inspection are then defined by national regulations or abattoir-specific work instructions. In Germany this is regulated by the General Administrative Regulation of Food Hygiene (AVV LmH), defining that at least lung, pleura, heart and liver must be inspected and documented by an abattoir-specific code number (BMEL, 2009). An alternative solution can be found in the Danish regulation, which stipulates nationally consistent code numbers to define every disease or lesion, and the medical condition concerned (CIR1H No. 9611, 2011). So, national scoring systems vary with regard to the respective disease definitions. These differences in scoring systems are a good illustration of the conclusions of Harley et al. (2012), saying that the main issue in meat inspection data is a non-standardized process at EU- and, furthermore, international level.

3.2. Lesion definition and recording methods

Many studies on meat inspection data of fattening pigs have analyzed estimated prevalence of recorded lesions. Prevalence is calculated as the proportion of individuals in a population who have a disease at a given time (Rothman, 2012). There is no approach in the literature that is considered as a gold standard for calculating the true prevalence (i.e. an error-free measurement of every single pig having disease at a given time). Instead, prevalence is estimated based on apparent and documented lesions. Since diseases will not be fully documented during meat inspection process, these crude estimates represent only a fraction of the true medical condition, and consequently result in an underestimated prevalence (Bonde et al., 2010; Enøe et al., 2003). During the last few decades, the estimated prevalence of lung, pleura, heart and liver lesions has been repeatedly estimated in several studies, but study premises, study scores and study methods have varied substantially. Nevertheless, an overview of the main results published in the literature of the last seven years is given in Table 1. Noticeable are the different disease definitions and criteria chosen by the authors to record

apparent organ lesions, e.g. “discrete areas of inflammation” vs. “ventro-cranial lesions with pleura adherence between or at ventral border of lobes”, both used as criteria for mild pleuritis (Meyns et al., 2011; van Staaveren et al., 2016). Other studies have summarized different diseases such as pneumonia and pleuritis to respiratory disorders (Bonde et al., 2010). Jirawattanapong et al. (2010) aggregated different pathologies, for example pleuritis as chronic and acute fibrinous disease patterns. Van Staaveren et al. (2016) used an adapted version of the British Pig Health Scheme (BPHS) scale to differentiate between mild and severe pleuritis, while Meyns et al. (2011) and Merialdi et al. (2012) applied the Slaughterhouse Pleurisy Evaluation System (SPES). The latter consists of four different scales, which may allow a differentiated assessment but may also affect the ability to detect pleuritis. These different approaches cause substantial variations in estimated prevalence, due to methodical differences as well as due to variation in disease definitions.

3.3. Current evaluation and feedback of meat inspection data

Abattoir data is potentially valuable for improving health at farm level by providing periodical feedback to the farmer on prevalence estimated on apparent lesions (Blaha, 2008; Breitenbuch and Blaha, 2005). Nevertheless, EC Directive No. 854/2004 does not stipulate the transfer and evaluation of meat inspection data to the farm of production, but several national regulations and organizations include the feedback as additional information in the abattoir settlement (BMEL, 2009; Agriculture and Food Council, 2013; Food Control Consultants, 2015). However, the evaluation of meat inspection data at farm level is difficult as several effects bias the data (see Section 4), thus leading to an underestimation of prevalence (Alban et al., 2015; Bonde et al., 2010; Teixeira et al., 2016). Another general issue with abattoir feedback is the comparability between abattoirs. Pigs originating from one farm are often delivered to several abattoirs. Lesions recorded at different abattoirs are influenced by abattoir-related factors, and thus the estimated prevalence sometimes varies within a batch of pigs. According to Blaha (2008) and Heinonen et al. (2018) a comparison of feedback lesions can only be informative if the data is sourced from one abattoir. Currently, the interpretation and utilization of reported lesions at farm-level depends on the knowledge of the farmer and/or the herd veterinarian (Devitt et al., 2015; Dickhaus et al., 2009).

4. Factors influencing lesion prevalence

In the literature, several factors influencing the estimated prevalence are mentioned and need to be considered when working with meat inspection data. Such factors can be included in statistical models. However, the characteristics and structure of factors in the available data need to be defined and circumstantiated first.

4.1. The effect of the meat inspector

Several studies showed a consensus view that meat inspection data, more specifically lesion recording, strongly depends on the meat inspector carrying out the examination (Alban et al., 2015; Bonde et al., 2010; Eckhardt et al., 2009; Stärk et al., 2014; Thomas-Bachli et al., 2012). During a 22-month period, Eckhardt et al. (2009) observed the assessment of thirteen veterinarians and calculated the repeatability of their findings. In total, 152,520 pigs were inspected and seven of the veterinarians showed good repeatability, whereas six showed significant differences during the study period. They concluded that a comparison of the recorded lesions among the veterinarians, and consequently a comparison between abattoirs was not possible, as variations in the assessment was shown. Most studies agree that sophisticated scoring systems may improve data quality, but the performance of the examination still depends substantially on the inspector. For example, Stärk et al. (2014) and Eckhardt et al. (2009) mentioned that

Table 1
Published disease prevalence of lung, pleura, heart and liver lesions calculated on meat inspection data of fattener pigs (2010–2017).

Organ/disease	Authors's criteria for disease finding	Prevalence (%)	Prevalence range (%)	Number abattoir	Number pigs	Production system	Period	Reference
Lung								
Pneumonia	Catarrhal pneumonia, acute haemorrhagic-necrotizing pneumonia, granulomatous pneumonia	43.4	–	1	968	C	5 m	Jirawattanapong et al. (2010)
Respiratory disorder	Pneumonia, pleuritis (acute and chronic)	42.0	–	1	2155	C	4 d	Bonde et al. (2010)
Respiratory disorder	Pneumonia, pleuritis (acute and chronic)	16.0	–	1	899	O	4 d	Bonde et al. (2010)
Pneumonia mild	<10% lobular modifications	13.8	3.2–26.8	1	247,505	C	39 m	Schleicher et al. (2013)
Pneumonia moderate	11–30% lobular modifications	11.9	4.8–23.5	1	247,505	C	39 m	Schleicher et al. (2013)
Pneumonia severe	>30% coalescent lobular modifications	4.7	1.5–16.2	1	247,505	C	39 m	Schleicher et al. (2013)
Chronic pneumonia	Not available	0.3	–	1	1173,213	C	12 m	Alban et al. (2015)
Chronic pneumonia	Not available	0.35	–	1	201,160	O	12 m	Alban et al. (2015)
Pneumonia	Plum-colored consolidation of lung parenchyma	10.9	–	1	3143	C	7 d	Teixeira et al. (2016)
Enzootic pneumonia-like lesions	Madec score $\geq 5/24$	9.8	–	1	72,700	C	12 m	Scollo et al. (2017)
Pleura								
Pleuritis	Chronic pleuritis & acute fibrinous pleuritis	45.4	–	1	968	C	5 m	Jirawattanapong et al. (2010)
Pleuritis	SPES score > 1	20.76	–	10	6335	C	4 m	Meyns et al. (2011)
Pleuritis mild	SPES score 1; ventro-cranial lesion with pleura adherence between or at ventral border of lobes	42.3	–	10	6335	C	4 m	Meyns et al. (2011)
Pleuritis moderate I	SPES score 2; dorso-caudal unilateral focal lesion	11.4	–	10	6335	C	4 m	Meyns et al. (2011)
Pleuritis severe I	SPES score 3; bilateral score 2 lesion of extensive unilateral lesion ($\geq 1/3$ of diaphragmatic lobe)	7.9	–	10	6335	C	4 m	Meyns et al. (2011)
Pleuritis severe II	SPES score 4; severe, extensive bilateral lesion ($\geq 1/3$ of both diaphragmatic lobes)	0.7	–	10	6335	C	4 m	Meyns et al. (2011)
Chronic pleuritis	SPES score ≥ 1	47.5	–	6	4889	C	3 m	Meriardi et al. (2012)
Dorsocaudal pleuritis	SPES score ≥ 2	25.1	–	6	4889	C	3 m	Meriardi et al. (2012)
Visceral pleuritis	Not available	14.1	4.9–22.2	1	247,505	C	39 m	Schleicher et al. (2013)
Parietal pleuritis mild	Not available	7.3	3.6–13.2	1	247,505	C	39 m	Schleicher et al. (2013)
Parietal pleuritis severe	Not available	8.3	3.9–12.0	1	247,505	C	39 m	Schleicher et al. (2013)
Chronic pleuritis	Not available	23.94	–	1	1173,213	C	12 m	Alban et al. (2015)
Pleuritis	Not available	19.06	–	1	201,160	O	12 m	Alban et al. (2015)
Pleuritis mild	Fibrotic adhesions to thoracic wall; fibrosis of pleura membrane	13.9	–	1	3143	C	7 d	Teixeira et al. (2016)
Pleuritis severe	BPFS scale 1; discrete areas of inflammation of the pleura	21.9	1.0–38.8	1	5628	C	5 d	Van Staaveren et al. (2016)
	BPFS scale 2; inclusive lungs attached to chest wall; whole lobes of the lungs	19.4	3.1–44.0	1	5628	C	5 d	Van Staaveren et al. (2016)
Pleuritis severe	SPES score ≥ 3	18.5	–	1	72,700	C	12 m	Scollo et al. (2017)
Heart								
Heart disorders	e.g. Pericarditis (further definition not available)	9.0	–	1	2155	C	4 d	Bonde et al. (2010)
Heart disorders	e.g. Pericarditis (further definition not available)	5.0	–	1	899	O	4 d	Bonde et al. (2010)
Pericarditis	Not available	3.1	2.5–3.9	1	247,505	C	39 m	Schleicher et al. (2013)
Chronic pericarditis	Fibrosis of pericarditis sac, with or without fluid	1.32	–	1	1173,213	C	12 m	Alban et al. (2015)
Chronic pericarditis	Not available	1.67	–	1	201,160	O	12 m	Alban et al. (2015)
Pericarditis	Not available	2.7	–	1	3143	C	7 d	Teixeira et al. (2016)
Liver								
Parasitic disorder	liver cirrhosis & milky spots in liver caused by migrating parasites	5.0	–	1	2155	C	4 d	Bonde et al. (2010)
Parasitic disorder	liver cirrhosis & milky spots in liver caused by migrating parasites	51.0	–	1	899	O	4 d	Bonde et al. (2010)
Milksports mild	Blaha score < 3	2.5	1.0–3.9	1	247,505	C	39 m	Schleicher et al. (2013)
Milksports severe	Blaha score > 3	4.7	2.0–10.1	1	247,505	C	39 m	Schleicher et al. (2013)
Milksports severe	Not available	3.0	–	1	1173,213	C	12 m	Alban et al. (2015)
Milksports severe	Not available	0.9	–	1	201,160	O	12 m	Alban et al. (2015)
Ascariasis	multifocal fibrotic lesions in liver stroma	5.7	–	1	3143	C	7 d	Teixeira et al. (2016)
Milksports severe	Number of milk spots lesions due to <i>A. suum</i> ; score 3 (> 10 lesions)	9.5	–	1	72,700	C	12 m	Scollo et al. (2017)

(continued on next page)

Table 1 (continued)

Organ/disease	Authors's criteria for disease finding	Prevalence (%)	Prevalence range (%)	Number abattoir	Number pigs	Production system	Period	Reference
Lung								
Milkspots total	Number of milk spots lesions due to <i>A. sumi</i> ; score 2 and 3 (≥ 4 lesions)	23.9	-	1	72,700	C	12 m	Scollo et al. (2017)

BPHS = British Pig Health Scheme; SPES = Slaughterhouse Pleurisy Evaluation System; C = conventional; O = organic; d = days; m = month.

motivation and dedication affect the individual workflow, and the experience of each inspector is probably most decisive for the detection rate.

By focusing on the agreement of organ finding assessments, only a few studies have investigated inter- and intra-observer reliability during meat inspection so far (Bonde et al., 2010; Eckhardt et al., 2009; Hoischen-Taubner et al., 2011; Schleicher et al., 2013). For example, Schleicher et al. (2013) compared the detection rates of twelve meat inspectors over three-and-a-half years by using the variance partitioning coefficient (VPC), which expresses the percentage of variation explained by the meat inspector (a high VPC represents large variation among meat inspectors). Extending the models by fixed effects (time, farm type, piglet producer), the authors determined a high VPC for skin lesions (3.93–20.77%) and hepatitis (2.92–18.87%), a moderate VPC for pneumonia (0.68–2.32%) and milk spots (0.02–1.96%) and a low VPC for pericarditis (0.01–0.14%), peritonitis (0.02–0.60%) and arthritis (0.00–0.44%). High variations in the observation of skin lesions, moderate variations in pneumonia and pleuritis and low variations in pericarditis have been confirmed by Eckhardt et al. (2009), Enøe et al. (2003) and Hoischen-Taubner et al. (2011). However, Hoischen-Taubner et al. (2011) found high variations in pleuritis and milk spotted liver. Non-standardized disease definitions seem causative for this high variation among recorded lesions (Eckhardt et al., 2009; Thomas-Bachli et al., 2012). Schleicher et al. (2013) assumed that skin lesions have been documented synonymously with scabies to explain the high variation. Furthermore, Hoischen-Taubner et al. (2011) concluded that meat inspectors used different thresholds for disease severity during the observation. Some recorded organ lesions, therefore, show a better reliability than others. In addition to the differences among various organs, high variations among meat inspectors occur in recorded lesions with several degrees of severity, too. For example, Schleicher et al. (2013) showed high ranges of mild (3.2–26.8%), moderate (4.8–23.5%) and severe (1.5–16.2%) estimated pneumonia prevalence. Moreover, severe disease lesions seem easier to diagnose because of clearly visible and definable symptoms, whereas differences between mild and moderate cases are often harder to distinguish. The lack of a standardized meat inspection process as well as explicitly defined disease definitions are the main causes of the low reliability sometimes observed (Enøe et al., 2003; Harley et al., 2012; Steinmann et al., 2014).

4.2. The effect of the abattoir

Accompanying the subjective assessment of the individual meat inspectors, several authors have reported extensive variations among abattoirs (Elbers et al., 1992; Enøe et al., 2003; Scollo et al., 2017; Thomas-Bachli et al., 2012; Vial and Reist, 2014). Cleveland-Nielsen et al. (2002), for example, established a multivariable model using meat inspection data of 18 different abattoirs to study several factors influencing the estimated prevalence of chronic pleuritis. They concluded that 19% of the variation of prevalence could be attributed to an abattoir effect.

The main factors that impede an unconstrained and adequate examination of the carcass are indicated in several studies, in particular the slaughter lineout, its line speed and light (Correia-Gomes et al., 2016; Scollo et al., 2017; Stärk et al., 2014; Vial and Reist, 2014; Harley et al., 2012; Thomas-Bachli et al., 2012). Thomas-Bachli et al. (2012) examined partial condemnation rates for nephritis and pneumonia lesions over six years in more than a hundred abattoirs in a Canadian province. Besides different condemnation rates between abattoirs, the authors found generally lower organ condemnation rates in large abattoirs and assumed inadequate examination because of a fast processing speed, too. The line speed is also indicated by Vial and Reist (2014), who assessed that less time for inspection caused condemnations even when some parts were fit for human consumption. Thomas-Bachli et al. (2012) also reported an effect of the abattoirs'

equipment, for example, the scalding tank water might contaminate the lung tissue and other lung lesions, so the lung can no longer be evaluated consistently. Besides the process design, work organization at the abattoir may also affect the detection of actual lesions. According to Blaha (1993) and Stärk et al. (2014), a sufficient number of inspectors should be present at the line to prevent a heavy workload. Therefore it may be concluded that the abattoir, as it presents the workplace of the meat inspectors, might affect the individual skills of inspectors and thus lead to inter-abattoir variations (Scollo et al., 2017; Thomas-Bachli et al., 2012).

4.3. The effect of the seasons

Apart from variations related to meat inspectors or the production process, several authors have indicated natural variations caused by seasonal effects (Kongsted and Sørensen, 2017; Schleicher et al., 2013; Scollo et al., 2017). During a 12-month study period in two abattoirs in the Netherlands (examined pigs = 552,152), Elbers et al. (1992) found a high estimated prevalence of pneumonia and skin lesions in the winter months, a higher percentage of condemned livers during autumn and high prevalence of pleuritis during summertime. Similar results for an estimated prevalence of pneumonia have been found by Scollo et al. (2017) during a 12-month study period in an Italian abattoir and a sample of 72,700 examined pigs. However, in contrast to Elbers et al. (1992) the authors estimated a higher percentage of condemned livers in spring and a higher estimated prevalence of pleuritis during wintertime. According to Elbers et al. (1992), a higher percentage of condemned livers during autumn might be caused by the seasonal development of *Ascaris suum* eggs. Variations in the estimated prevalence of milk spot lesions might be caused by the different regions where the data was collected. Taking into account the optimal temperature of 15 °C for larval development (Nansen and Roepstorff, 1999), a seasonal peak in the estimated prevalence of condemned livers occurs earlier in the south (e.g. Italy) than in the north (e.g. the Netherlands) because of different climatic conditions (Scollo et al., 2017). Similar seasonal climatic variations were mentioned by Thomas-Bachli et al. (2012) when comparing different abattoirs in a province of Canada. The authors explained different kidney condemnation rates in different regions to be caused by climatic variations. However, a significant association between region and season could not be found in the lung condemnation rate. Eckhardt et al. (2009) were not able to find any significant seasonality in lung lesions or pneumonia during their 22-month period. Nonetheless, season has an effect on meat inspection data and may result in the under-/overestimation of prevalence if not considered.

4.4. The effect of the farm

Meat inspection data and hence the prevalence of organ findings is affected by farm of origin on several different levels. Some studies have noticed differences in prevalence among pig herds (Elbers et al., 1992) and also an associations between an increased incidence of recorded lesions and poor health and welfare on the farm of production (Teixeira et al., 2016). Other studies have compared meat inspection findings between different production systems, i.e. conventional and free-range housing systems (Alban et al., 2015; Bonde et al., 2010; Ebke and Sundrum, 2005; Kongsted and Sørensen, 2017). Among others, Kongsted and Sørensen (2017) found a higher estimated prevalence of milk spots, tail and skin lesions in free-range than in conventional systems. Higher exposure to climatic change and exposure to insects in free-range systems were assumed to be possible reasons. The authors detected a large influence of the farm of origin in relation to milk spots, tail lesions and airway infections (intraclass correlation coefficient 21–35%). Alban et al. (2015) and Ebke and Sundrum (2005) examined the prevalence based on the calculation of apparent lesions in pigs of different housing systems. The latter study compared lung and liver

lesions of 46,411 conventional and 4,132 organic pigs and found a higher occurrence of mild lung lesions in conventional pigs (47.2% vs. 43.7%) and a higher estimated prevalence of milk spots among organically raised pigs (57.0% vs. 35.8%).

Besides variable production systems and management conditions, the herd health status of pigs influences pigs' performance (Heinonen et al., 2001) and should be considered when analyzing meat inspection results. Several studies have classified the herd health status of pigs and added this factor to their models (Enøe et al., 2003; Heinonen et al., 2001; Petersen et al., 2002). Herds with a higher health-status showed a lower number of apparent lesions during meat inspection than herds with standard or no classification (Enøe et al., 2003; Heinonen et al., 2001).

To estimate the amount of variation in meat inspection findings at farm level, Schleicher et al. (2013) used the VPC based on logistic multilevel models with cross-classified random effects. The estimated VPC showed a large variation among the farms for mild and severe milk spots (0.26–17.4% and 0.12–43.5%), suggesting that up to 43% of the variation in severe milk spots can be attributed to the farm. As pneumonia and pleuritis might also be strongly affected by the farm, the recording of other lesions such as pericarditis, arthritis and lung adhesion carcass exhibited only low VPC estimates. Further studies have included a farm-effect in their statistical models, with for example Scollo et al. (2017) reporting that this effect explains 36–68% of the variation in acute lesions (i.e., lesions of the lung, pleura and liver). Referring to the prevalence of chronic pleuritis, Cleveland-Nielsen et al. (2002) attributed 71% of its variation to the herd of origin.

5. Using meat inspection data for breeding purposes

5.1. Validity of recorded lesions

The validity of recorded lesions to monitor animal health at farm level has been examined and discussed in the literature since the late 70s (Bäckström and Bremer, 1978). EFSA (2011) indicates that meat inspection data might be a useful adjunct to provide information on animal health, and that it could contribute to an extended scoring system for health characteristics. However, as described in the previous sections, the examination and documentation of organ lesions contain several challenges that need to be considered when aiming to use recorded lesions for animal breeding. Nevertheless, literature provides substantial indications supporting that recorded lesions during meat inspection correlate with on-farm health data. Several studies examined the association and agreement between these two data pools (Correia-Gomes et al., 2016; Dickhaus et al., 2009; Knage-Rasmussen et al., 2015; Teixeira and Boyle, 2014; van Staaveren et al., 2017). Dickhaus et al. (2009) estimated a positive correlation between a herd health score, which included among others the frequency of recorded lesions of previous meat inspections, and on-farm health indicators (correlation = 0.80). In a similar study, Van Staaveren et al. (2017) documented moderate correlations between the estimated prevalence of on-farm health indicators and recorded lesions during meat inspection (bursitis and severe tail lesions). The authors suggested that lesions, recorded during meat inspection, have value to assess welfare on farm. However, they reported a high variation (9–45%) in the estimated prevalence of on-farm health indicators. Besides excluded effects, such as for example on-farm management and the disease status of the pigs, this variation could result from the snapshot assessment of the study. Furthermore, the authors admitted that lesions recorded at the abattoir were not necessarily from the same pig that was observed at the farm. Knage-Rasmussen et al. (2015) could not find any association between on-farm measures and meat inspection data.

The validity (of meat inspection) is determined by the sensitivity (SE), which denotes the proportion of true positives, and the specificity (SP) of recorded lesions, which represents correctly identified negatives

(Altman and Bland, 1994). Several studies have reported a variable, estimated prevalence of recorded lesions over time (Enøe et al., 2003; Kongsted and Sørensen, 2017; Scollo et al., 2017; Steinmann et al., 2014; Thomas-Bachli et al., 2012). Assuming that variability in estimated disease prevalence might be affected by changes in the SE and SP of the examination, Enøe et al. (2003) assessed the validity of a traditional and an extended meat inspection for the detection of chronic pleuritis during two periods of 12 months each. The authors used the approach by Hui and Walter (1980) to estimate SE and SP capable of estimating the error rates of a diagnostic test without a gold standard. The SE of traditional meat inspection showed large variations between the four abattoirs (period I: 28.8–61.4%, period II: 39.2–87.3%), whereas a higher SE (period I: 85.7–94.9%, period II: 73.9–93.0%) was shown in the extended inspection. According to the authors, an observed increase in prevalence during period II might have been caused by an increase in the SE of the meat inspection. Consequently, biased recording leads to high variations in estimated prevalence among abattoirs, which is also exhibited in Table 1. Therefore, the estimates of prevalence, SE and SP must be recognized as abattoir- and disease-specific (Blaha, 2008; Bonde et al., 2010; Enøe et al., 2003; Stärk et al., 2014) and the validity should be evaluated regularly (Hoischen-Taubner et al., 2011).

5.2. Initial study estimated heritability of organ lesions

Despite the obstacles described, the first study of recorded lesions as a potential data pool to improve pig health was carried out by Mathur et al. (2018). The meat inspection data of 140,375 pigs in different abattoirs were recorded during a five-year period. The authors estimated heritabilities for pericarditis (0.14), pleuritis (0.09), joint disorders including bursitis (0.17) and liver lesions (0.24). This study demonstrates the capacity of using the data on recorded lesions to improve pig health, even though no demanding data treatment was applied. Neither did Mathur et al. (2018) indicated how many abattoirs participated in the study, nor did the authors provided further information about what adjustments were applied to account for meat inspector or abattoir effects. This raises the question of to what extent these results are generalizable. Although the general willingness of breeding organizations to utilize this data pool seems limited at present, these initial results indicate the potential of meat inspection data for breeding purposes.

5.3. Potentials and limitations of recorded lesions for breeding purposes

The application of health-related phenotypes based on meat inspection data is currently not common in breeding programs, mainly because of two factors: first, as already described the quality of data recording and second the data structure. Regarding to data quality, non-standardized processes expose high-scale effects and heteroscedasticity between abattoirs, resulting in differences in means or standard deviations of the calculated prevalence. The standardization of the meat inspection process at least at national level is necessary (Harley et al., 2012) to enable the measurement of standardized phenotypes (Haas et al., 2012). Additional circumstances, such as missing records of animals that died during the fattening period (Harley et al., 2012) and the consideration of lesions caused during transport or due to the slaughter process (Carroll et al., 2016; EFSA, 2012; van Staaveren et al., 2015), are not currently considered. Furthermore, there is no standardized approach in the literature to estimate prevalence, which is a substantial challenge when aiming for the use of recorded lesions as phenotypes.

Other issues that limit the usage of recorded lesions for breeding purposes are the validity and reliability of meat inspection data. Conflicting results of studies, examining whether meat inspection data monitor on-farm health data, may be the result from different datasets, variations in the time and quantity of observation, different measures

and diverging statistical methods. There is a need for more comparable studies to examine what kind of on-farm database shows an association to recorded lesions during meat inspection. It should be additionally investigated whether every (indirectly) measured organ of both databases is comparable or whether it is limited to only a few that show high correlations. Besides validity, high variations within abattoirs occurring in organ specific prevalence (e.g. lung and pleura lesions) indicate the low reliability of the examination. A large proportion of the variance is therefore traced back to measurement errors, thus random and systematic errors. As a result, the reliability of the meat inspection examination is low. Low reliability in recorded lesions, however, can be considered as a manageable issue, because meat inspection data generally provide a high number of cases that may compensate for the low reliability of subsequent statistical evaluations.

The improvement of meat inspection reliability is discussed in the literature to lower inter-abattoir variation. Blaha (1994), for example, suggested a scoring system to minimize inter-observer variation by recording differences in the severities of respiratory diseases. This approach is currently being applied in a modified version by the German AVV LmH. Additionally, Steinmann et al. (2014) created a surface-related evaluation system to standardize and to improve the reliability of the documentation of lung lesions. After an individual, two-week training course for the official auxiliaries, the authors compared the application among a group of official auxiliaries and an official veterinarian. The deviations between the detection rates for moderate lung lesions of the two inspector groups decreased from 75% to 15%. However, even using this validated and standardized scheme, deviations in the detection rates between the inspectors were still higher than 10%. Besides personal training, automation of meat inspection seems appropriate to be considered for the interrater effect. The first studies on automating meat inspection were carried out by Blömke and Kemper (2017), who implemented 2D video-monitoring to record ear and tail lesions in the carcass. The monitoring system showed a high level of accuracy in the detection and recording of ear (SE 75–80%, SP 94–97%) and tail lesions (SE 73–92%, SP 99–100%). However, the standardization of disease definitions and thresholds remain challenging.

At present, automatic recording of organ lesions during meat inspection is not widely applied in abattoirs, thus, comprehensive and regular training is often cited as an adequate approach to improve the inspectors' experience and hence their performance (Luukkanen et al., 2017; Scollo et al., 2017; Stärk et al., 2014; Steinmann et al., 2014; Thomas-Bachli et al., 2012). Supporting evaluation systems, like the system of surface-related lung lesions by Steinmann et al. (2014), might also improve data recording by simplification of the scoring, especially in the existing practice and production environment at the abattoir. Unfortunately, no or insufficient incentives for such practice are currently given. Alternatively, the raw data could be adjusted retrospectively, for the meat inspector effect by identifying every inspector carrying out the examination. However, challenges in data protection and personal rights currently prohibit such an approach. Statistical pre-processing, e.g. by using linear mixed-models or robust data transformations, therefore, seem valuable for further examination. Such approaches might be able to account for meat inspector and day of slaughter effects in the data. Besides this, farm-specific information (e.g. herd-health status, housing system), could be linked to recorded lesions, to account for it in subsequent statistical evaluation. Nevertheless, further research is needed whether and to what extend factors at animal-, housing- and management-level are responsible for differences in apparent lesions among different farms.

In addition to the data quality, there are also requirements in data structure to improve the accuracy of estimated prevalence. Pigs are not usually delivered by the farmer to the abattoir as a batch. Instead, the abattoir receives a mixed group of pigs originating from different farms from a pig-trader. Reported lesions, in this case, can sometimes be modified by the trader. Additionally to the data modification, abattoir-

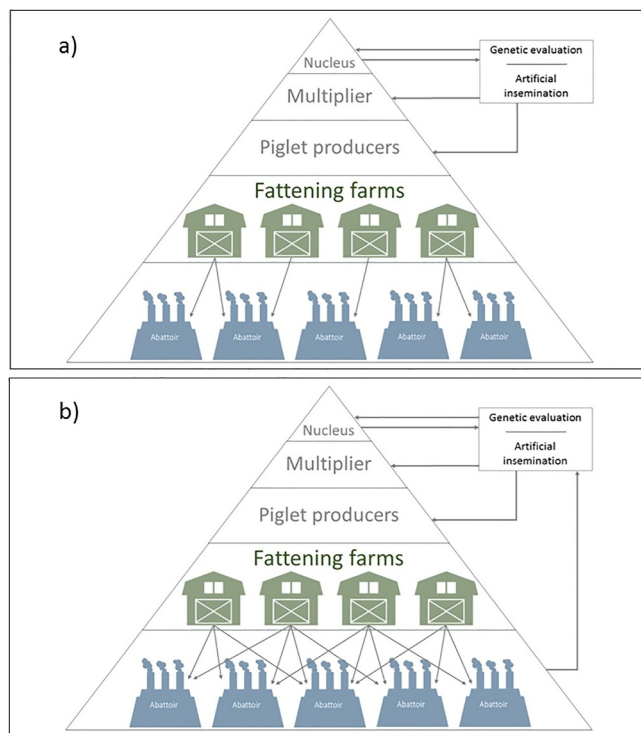


Fig. 1. Current (a) and desirable (b) data structure between farm- and abattoir-level.

and trader-specific code numbers to describe apparent lesions hinder the merger of meat inspection data from different abattoirs. This in turn hampers the preparation of meat inspection data for the purpose of evaluation and thus the use for breeding. Nationally consistent code numbers for recorded lesions might simplify data management, and variations resulting from human interaction could be minimized. As mentioned previously, there is no incentive for abattoirs to improve the meat inspection process, because efforts to provide this additional service would not be profitable. Breeding organizations, however, might find this data pool to be highly valuable, and could negotiate cooperation agreements with abattoirs to generate a well-linked data structure and thus high quality data collection. In conventional breeding schemes, a nucleus animal is tested under practical conditions in several contract farms (e.g. by using artificial insemination). The performance of the relatives is used, in turn, to select nucleus animals (Knap, 2005; Turner et al., 2018). Due to this organized mating program, a well-linked data structure is generated in the multiplier and piglet producer level, and biasing environmental factors of the performance-data will be adjusted. A similar approach would be desirable to connect farm and abattoir level. A standard procedure to adjust for the effects discussed in the previous sections is a linear model approach. However, a well-linked data connection between abattoirs would also greatly improve the performance of such an approach. Currently, pigs originating from a farm are usually sent to one abattoir, and just a few farmers deliver their pigs to more than two abattoirs (Fig. 1a), thus the adjustment of the data is not possible in this case. In a desirable data structure, relatives of one nucleus boar that are fattened at one farm are slaughtered and inspected in various, different abattoirs (Fig. 1b). Additional recording of ear tags and marking stamps at the abattoir is needed to link abattoir records with pedigree data. The implementation and data transfer to generate such a data structure requires a high degree of organizational effort.

No simple solution exists to instantly improve the raw data and optimizations at many places are therefore necessary to enable the measurement of standardized phenotypes.

6. Conclusions

Consistently documented organ lesions during meat inspection might prove to be a valuable data pool of phenotypes for breeding based on health traits. However, the reliability and thus the validity of several lesion recordings is currently inadequate. A comparison of abattoirs is possible to a limited extend only, because variations arise both within (inter) and between (intra) abattoirs. There are many possible reasons (standardization, meat inspector), as shown in this review, but no simple solution exists to immediately improve the raw data. Optimizations at many places (data quality and structure) are therefore necessary, even if they seem challenging to implement. The potential of recorded lesions is nevertheless valuable for animal-health and breeding purposes in the future.

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